

► Introduction

In August 2023, the ILO published the first global employment estimates of potential occupational exposure to generative artificial intelligence (GenAI) (Gmyrek et al., 2023). This research was prompted by concerns – that have featured prominently in the media – of the potential for the technology to replace large shares of knowledge work.

Since the release of the 2023 estimates, there have been further technological advances in GenAI, including the introduction of Large Language Models (LLMs) with expanded capabilities in image recognition and audio and video production. Moreover, many new GenAI applications have been developed as a result of the possibility of linking GenAI models with Application Programming Interface (API) tools. Indeed, GenAI has been likened to a general-purpose technology, meaning that it may continue generating new applications and innovations across various sectors, driving long-term economic and societal transformations.

Despite this rapidly developing landscape, it is important for policy makers to develop deeper understanding of how technological advances might impact existing jobs. To address this need, the Research department of the ILO has partnered with NASK, the National Research Institute of the Ministry of Digital Affairs in Poland, to develop a more precise assessment of potential effects of GenAI on employment across countries.

This study builds on the method developed by the ILO (Gmyrek, Berg and Bescond, 2023 - GBB hereafter), which used task descriptions from the 4-digit International Standard Classification of Occupations (ISCO-08) and the GPT-4 model to estimate task-level and occupational scores of exposure to GenAI technology and then derives global employment estimates based on official ILO labour force data. But the study includes several key refinements.

The first step in our analysis is based on Poland's national 6-digit classification of occupations that includes nearly 30,000 tasks. This expands tenfold the number of tasks given in the ISCO-08 structure and enables a fine-grained assessment of tasks' automation potential, based on a more up-to-date set of occupations and tasks than those captured in the ISCO-08 system. As Poland's 6-digit classification is aligned with the 4-digit ISCO-08, the occupational scores can be compiled at the 4-digit level allowing a straightforward comparison and updating of the 2023 estimates of the ILO (GBB). In addition, the study improves the methodological framework of GBB, by drawing on the combined strengths of human and AI abilities, with an additional layer of expert verification.

Specifically, we surveyed 1,640 people currently employed in Poland in each ISCO-08 1-digit group to rank the automation potential of a representative sample of tasks that pertain to their occupational group. As the second step, we asked a mix of national and international experts to conduct a detailed review of a large sample of tasks previously assessed in the survey to validate or adjust the level of automation scores through an iterative process. Since we maintain the hierarchical link to ISCO-08 and focus on technological feasibility rather than specific country context, by combining opinions of local and international experts, we create a system that enables more precise predictions of the potential of automation of individual occupational tasks with GenAI technology, including at the international level. To achieve that, we input the human scores into an AI model, to generate scores reflecting human judgement for nearly 30,000 individual tasks pertaining to 2,500 occupations at 6-digit level in Poland. We then use the same AI predictor to re-generate scores for the tasks in ISCO-08, previously provided by GBB (2023), thereby adjusting the 2023 index of the ILO, which now benefits from both the input of people

► 1 Task-based approaches to automation in the literature

Central to the study of technology on work is the insight that jobs are a “bundle of tasks” (Autor, 2015). As such, task automation might, or might not lead to job automation, depending on the importance of a particular task to an occupation. Analyses of employment effects of automation that use the task-based approach, including this study, attempt to capture two central considerations: (i) the ability of a specific technology to automate a given task and (ii) the character of existing occupations in the labour market, based on the detailed tasks performed by these occupations. By modelling the intersection of these two dimensions, one arrives at a range of possible projections of how such technology could interact with occupational tasks, leading either to their full automation through technology-driven substitution, or to a transformation, through partial automation and human-machine complementarity. While the task-based framework certainly has limitations, recent academic literature shows the analytical advantages of this method for modelling employment and labour market impacts (Acemoglu et al., 2024; Nurski and Ruer, 2024).

Research on AI and tasks has shown important improvements in methods and scope in recent years, which includes using AI tools for research purposes. In one of the early papers on Machine Learning (ML) and jobs, Frey and Osborne, (2013) fed the opinions of a group of IT technology experts into an ML algorithm to develop occupational-level projections of automation potential for the US job market. The emergence of GenAI APIs has simplified the possibility of combining human expertise and algorithmic capacities for research purposes. Relying on this idea, Eloundou et al. (2023) demonstrated a close alignment of GPT-4 predictions with a survey of 70 AI experts on the potential of automating occupational tasks with LLMs, and subsequently built on this feature to develop synthetic automation scores for the US market.

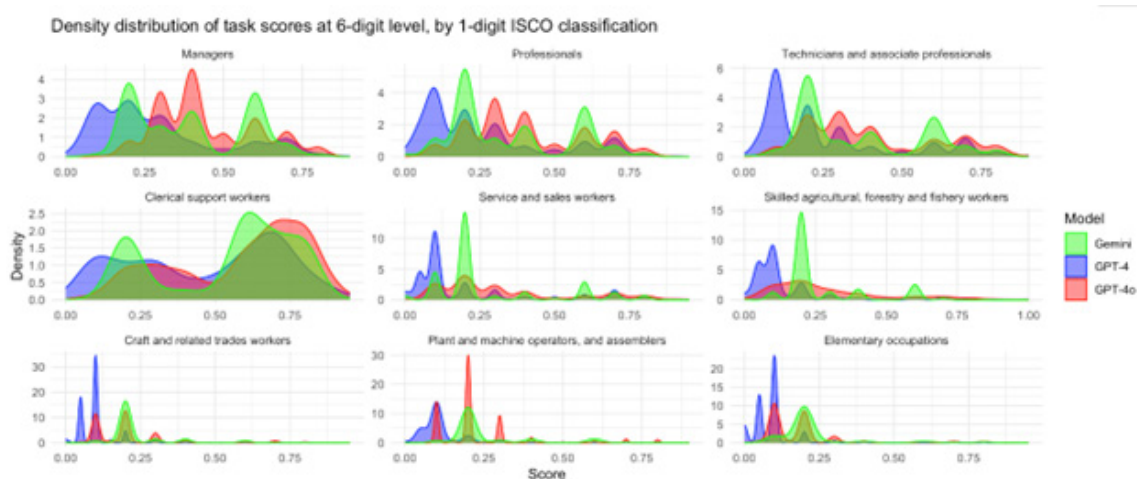
Most task-based studies are focussed on the US, due to the very detailed nature of the O*NET database, as well as its public accessibility. This typically forces researchers interested in the global picture or other countries and regions to base their analysis on the strong assumption of similarity of local economic and labour effects to those projected in the US context. In reality, occupational tasks can vary significantly among individual countries, which poses a particular challenge in developing countries. As such, working with tasks in the ISCO-08 system offers a useful and simple common denominator for projections across different countries, regions and income groups. Our method combines the strengths of this global framework with a significantly more detailed national classification system in Poland, which provides a detailed list of occupations and tasks, based on the latest labour market data. This helps us establish a starting point for modelling interaction of GenAI’s abilities with up-to-date tasks and occupations in a setting located between the countries with highest incomes and the emerging economies (Figure 1). Using a combination of GenAI tools, we subsequently transfer this knowledge back to the tasks in ISCO-08 system.

► 3 Assessment of tasks' automation potential

3.1. Initial Algorithmic predictions

As the first step, we conducted a purely algorithmic assessment of potential exposure to automation for each task in the Polish 6-digit system, relying on three advanced LLMs available on the market at the end of 2024: GPT-4, GPT-4o and Gemini Flash 1.5. The method relies on the approach previously used by GBB (2023), with a sequential API call devised to assign a score ranging from 0 to 1 for each individual task, where 0 means the task cannot be automated and 1 means the task could be fully automated with GenAI technology (the task can be conducted without participation of a human operator).⁴ The LLM is provided with information on the occupation to which the task pertains in ISCO-08 and requested to produce a justification for each predicted score.

► **Figure 2. Distribution of synthetic automation scores from 3 LLMs, by ISCO-08 1-digit**



The overall distribution of these synthetic automation scores is generally consistent with scores based on ISCO-08 documentation (GBB, 2023; Gmyrek, Winkler and Garganta, 2024) and O*NET (Elondou, 2023; Felten, 2023): the highest exposure to GenAI is typically observed among groups with the largest share of tasks of cognitive nature and it declines with tasks of physical nature, which are more prominent among occupational groups 4-9 (Figure 2). The largest density of high automation scores is observed among clerical workers and is stable across the three LLMs. In other groups, we observe some important differences in the LLMs' predictions. In particular, Gemini Flash 1.5 seems to provide a bimodal scoring, with most scores falling either below 0.25 or above 0.6. The other two models provide a wider range of scores, but still with important divergences, which makes it difficult to judge which model one should follow.⁵ Therefore, we do not examine these distributions in greater detail but proceed directly to improvements that could enhance coherence and stability of our predictions, regardless of the LLM applied.

⁴ In GBB (2023), the scores and justifications were assessed for consistency and stability of predictions over time, with the written justifications reviewed by humans. Tasks with scores above 0.8 (high possibility of automation) were transformed into embeddings, with a semantic clustering algorithm applied to identify the major groups of such tasks, which are subsequently reviewed by humans.

⁵ A comparison to GBB (2023) scores suggests the closest alignment with GPT-4o synthetic scoring of the Polish tasks' database.